# Yingrui Hou

## Simulation & Model Validation | Digital Twin Applications

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https://yingrui-hou.github.io/PersonalPage/

### Profile

Physics PhD with postdoctoral experience connecting measurements, simulation and calibrated models. Builds Python/C++/Geant4 workflows and uses TensorFlow/GPU numerical computing. Experience in real-data validation of learned models supports transfer to digital twins, simulation engineering and industrial R&D. Brings research project coordination, technical presentation and international collaboration.

### Core skills

- Simulation: Geant4/C++, detector geometry, interaction stepping, response logic, event-level outputs and measured response maps.
- Validation: simulation-to-measurement alignment, deep-learning calibration, image-recognition model checks, bias correction and uncertainty analysis.
- Signals & models: time-profile decomposition, likelihood fitting, statistical inference, CatBoost/gradient boosting and data-driven model correction.
- Computing: Python, C/C++, ROOT/RooFit, NumPy, pandas; TensorFlow/GPU model evaluation; Linux, Shell/YAML/CMake and shared-code review.

### Research experience

**Postdoctoral Researcher | 2021–2026**

CNRS — LAPP, Annecy (2021–2023); LPC/LPCA, Clermont-Ferrand (2023.12.01–2026.05.31). Research in LHCb and GRAiNITA collaborations.

- Built and adapted Geant4/C++ detector simulations, implementing geometry, stepping and analysis outputs to study optical and detector response.
- Connected beam-test measurements and position-dependent response maps to simulated energy deposits, evaluating performance against measured rather than ideal response.
- Developed time-resolved signal analysis and ROOT/C++ correction workflows; checked reconstructed energy before and after correction under the studied conditions.
- Used real data to evaluate and optimise the ECAL clustering model; investigated reconstruction discrepancies to guide calibration and software follow-up.

**Doctoral research training | 2016–2021**

University of Chinese Academy of Sciences — joint Master and PhD in Particle Physics. Training in quantitative problem formulation, scientific computation, experimental validation and communication of research results.

### Education

- Joint Master and PhD in Particle Physics — University of Chinese Academy of Sciences, 2016–2021.
- BSc in Applied Physics — China University of Mining and Technology, 2010–2014.

## Selected project evidence

### Measurement-informed detector model

Integrated measured beam-test response maps into Geant4 energy-deposition simulation and compared measured and idealised responses. Non-uniformity contributed below 1% to the constant term in the first GRAiNITA prototype study. Transfer: model calibration and validation for digital twins.

### Signal decomposition and correction

Extended simulated scintillation timing and fitted signal-component fractions to correct reconstructed energy. A representative rotation-based PSD study reduced a resolution term by about 2×; material comparisons exposed separation limits. Results apply to the studied configuration.

### Real-data validation of learned models

Calibrated and validated a deep-learning flavour tagger; used real data to check and optimise an image-recognition model for ECAL clustering. Connected model-performance findings to calibration and reconstruction investigations.

### TensorFlow/GPU scientific modelling

Constructed and evaluated complex-amplitude probability density models for a joint gamma measurement using TensorFlow and GPU acceleration. Applied numerical computing to a complex statistical model.

For flavour tagging and ECAL clustering, participated in model-construction and training discussions for downstream calibration and validation; did not train these two models.

## Research training applied to engineering

- Adaptability and learning: applied likelihood inference, CatBoost, TensorFlow/GPU and Geant4 to different research problems through literature, implementation and data checks.
- Research coordination: organised connected measurement, analysis and validation tasks with reusable components and documented assumptions.
- Presentation and teamwork: presented at GDR-InF 2022 and FCC/DRD France 2025, including on behalf of GRAiNITA; contributed to shared code and technical reviews.

## Languages & mobility

Chinese: native | English: C1 | French: B1–B2. Long-distance cycling and travel; open to relocation in Europe.

## Project references

- Measurement-informed detector model (https://yingrui-hou.github.io/PersonalPage/index.html#work-4)
- Signal decomposition and correction (https://yingrui-hou.github.io/PersonalPage/index.html#work-5)
- Real-data validation of learned models (https://yingrui-hou.github.io/PersonalPage/index.html#work-6)
- TensorFlow/GPU scientific modelling (https://yingrui-hou.github.io/PersonalPage/index.html#work-8)